LESSON 10.2

Math is Beauty

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use bilateral symmetry to describe structure in nature and in human creations.

WHY THIS MATTERS There are over 17,000 species of butterflies in the world, and most people are fascinated by them. Part of the fascination is mathematical: math creates beautiful designs, many of which are found naturally in plants and animals.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
symmetric *Spanish: simétrico*	Having two halves that match — one side is a mirror image of the other.	A butterfly: the left wing is the mirror of the right.
mirror image *Spanish: imagen reflejada*	A copy of a figure flipped across a line, so the two halves face each other.	Fold the layer down the middle: each leaf lands on its partner.
bilateral symmetry *Spanish: simetría bilateral*	Reflective symmetry in plants and animals — the two sides of the body mirror each other.	All 17,000+ butterfly species have it — two sides, one mirror line.
line of symmetry *Spanish: eje de simetría*	The line you can draw through a figure so each side is a reflection of the other.	The line straight down the butterfly's body.
balance *Spanish: equilibrio*	An even, steady arrangement — neither side feels heavier than the other.	Two matching halves — neither side feels heavier than the other.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I look at the two sides of the butterfly, and they look like mirror images — the butterfly is symmetric.

1. All butterflies have reflective symmetry. In plants and animals, this is called bilateral symmetry.
2. To show it, I draw a line down the middle of the butterfly's body: everything on the left is reflected on the right.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

This plant grows in layers, and each layer is made up of two leaves — so 3 layers hold 3 × 2 = 6 leaves.

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ has a line of symmetry _____ because the two halves are _____.
- I have seen symmetry in _____ .
- The two sides _____ .
- A _____ has bilateral symmetry.

Word bank symmetric | mirror image | bilateral symmetry | line of symmetry | balance

WATCH OUT!

A common mistake

Thinking a figure lacks bilateral symmetry just because its line of symmetry is not vertical

LESSON 10.2

Math is Beauty

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use bilateral symmetry to describe structure in nature and in human creations.

WHY THIS MATTERS There are over 17,000 species of butterflies in the world, and most people are fascinated by them. Part of the fascination is mathematical: math creates beautiful designs, many of which are found naturally in plants and animals.

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

What does it mean that a butterfly is symmetric?

(A) The butterfly has exactly two wings

(B) The two sides of the butterfly look like mirror images

(C) The butterfly's wings are very colorful

(D) The butterfly is larger than most insects

HINT Think about what you see on the left wing versus the right wing.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

In plants and animals, reflective symmetry is called:

(A) rotational symmetry

(B) a mirror image

(C) bilateral symmetry

(D) balance

HINT The word starts with 'bi-', meaning two.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A plant grows in layers, and each layer is made up of 2 leaves. How many leaves does the plant have after growing 4 layers?

- (A) 4 leaves
- (B) 6 leaves
- (C) 8 leaves
- (D) 16 leaves

HINT How many leaves are in ONE layer?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

How can you SHOW that a layer of two leaves has bilateral symmetry?

- (A) Count the total number of leaves on the plant
- (B) Measure how tall the plant is
- (C) Draw a line down the middle of the layer and check that each side mirrors the other
- (D) Color the two leaves different colors

HINT Symmetry is about two sides matching.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Answer in complete sentences.**

Where have you seen symmetry before? Name one place and describe it.

SENTENCE STARTER I have seen symmetry in ___ .
The two sides ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Name one animal besides a butterfly that has bilateral symmetry.

SENTENCE STARTER A ___ has bilateral symmetry.
Its line of symmetry runs ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Find something in this room that you think has bilateral symmetry and your partner thinks does not. Where exactly would the line of symmetry go, and what makes the two halves mirror images rather than just similar?

SENTENCE STARTER ___ has a line of symmetry ___ because the two halves are ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 10.2

Math is Beauty

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use bilateral symmetry to describe structure in nature and in human creations.

WHY THIS MATTERS There are over 17,000 species of butterflies in the world, and most people are fascinated by them. Part of the fascination is mathematical: math creates beautiful designs, many of which are found naturally in plants and animals.

PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

Why do people find symmetry pleasing?

(A) Our brains are wired to favor balance, and bilateral symmetry provides balance

(B) Symmetric things are always brightly colored

(C) Symmetry only appears in rare and expensive objects

(D) Symmetric shapes are always larger than non-symmetric ones

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

There are over 17,000 species of butterflies, and ALL butterflies have reflective symmetry. About how many of those species have reflective symmetry?

(A) About half of them

(B) Only the most colorful ones

(C) None — symmetry is only in plants

(D) All 17,000+ of them

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A plant grows 2 leaves at a time, one layer per week. How many leaves has it grown after 6 weeks?

(A) 6 leaves
(B) 8 leaves
(C) 12 leaves
(D) 36 leaves

PER 1

4 **Circle the letter of the best answer. Show how you know.**

Which is the best test for whether a figure has bilateral symmetry?

(A) Check whether the figure has an even number of parts
(B) Find a line where one side is the mirror image of the other
(C) Check whether the figure is found in nature
(D) Check whether the figure looks beautiful to you

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

Are the two leaves in a real plant layer EXACTLY symmetric? Why might this be?

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Where else do you see bilateral symmetry — in nature OR in something humans made?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Find something in this room that you think has bilateral symmetry and your partner thinks does not. Where exactly would the line of symmetry go, and what makes the two halves mirror images rather than just similar?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 10.2

Math is Beauty

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use bilateral symmetry to describe structure in nature and in human creations.

WHY THIS MATTERS There are over 17,000 species of butterflies in the world, and most people are fascinated by them. Part of the fascination is mathematical: math creates beautiful designs, many of which are found naturally in plants and animals.

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each transformation idea to its description.

_____ Reflection
_____ Rotation
_____ Translation
_____ Rotational symmetry

(A) Flip across a mirror line
(B) Turn around a center point
(C) Slide without turning or flipping
(D) Fits onto itself in less than a full turn

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Your friend insists that only ONE of two flowers has bilateral symmetry, but both pass the mirror-image test — just across lines pointing in different directions. What is the best response?

(A) Your friend is right, because a real line of symmetry must always be vertical
(B) Both flowers have bilateral symmetry — a line of symmetry does not have to point the same direction in every figure
(C) Neither flower has bilateral symmetry, because flowers are not animals
(D) Agree with your friend, because two people should not disagree about math

SHOW YOUR WORK

3 Circle the letter of the best answer. Show how you know.

A symmetric garden design puts the same planting on each side of a center path. One side uses 14 tulips and 9 daisies. How many flowers does the WHOLE symmetric design use?

(A) 23 flowers
(B) 28 flowers
(C) 46 flowers
(D) 126 flowers

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 Answer in complete sentences.

You and a friend are studying a building for an art project. The building's left half mirrors its right half. What can you say about the building in terms of bilateral symmetry?

WRITE YOUR ANSWER

5 Answer in complete sentences.

The lesson says math CREATES the beautiful designs we find in nature. Someone argues back: 'Nature is beautiful, but that has nothing to do with math.' Use the butterfly, the plant, or symmetry in your answer to respond.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Find something in this room that you think has bilateral symmetry and your partner thinks does not. Where exactly would the line of symmetry go, and what makes the two halves mirror images rather than just similar?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help